

Jason Wang

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WORK EXPERIENCE

BC Technologies Corp.

2022 - 2023

Apprentice

Remote

- Maintained production website (FutureSphere Academy) using the stack of NextJS, PostgreSQL, and TailwindCSS.
- Worked in an agile development methodology environment.

EDUCATION

Harrisburg University of Science & Technology

June 2027

Bachelor of Science in Computer Information Science

Harrisburg, PA

- **GPA: 3.86/4.0**
- **Data Structures, Essential Algorithms, System Information Design**
- **Operating Systems, Programming Language Concepts, Formal Language & Automata Theory**

Adams County Technical Institute

June 2024

Computer Networking

Gettysburg, PA

- **GPA: 3.9/4.0**
- **1st in Competitive Programming at SkillsUSA (District 13)**
- **Special Recognition for 2023 Congressional App Challenge**

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** CompTIA A+
- **Technologies:** Python, JavaScript, TypeScript, Java, C, C++, React, NextJS, HTML, CSS, TailwindCSS, SQL, PostgreSQL, MariaDB, SQLite, Convex, ClerkJS, Unix, Windows, Docker, Claude Code
- **Skills:** Problem Solving, Strategic Planning, Quick Learner, Self Motivated, Organization, Teamwork
- **Languages:** English, Mandarin Chinese
- **Interests:** Gaming, Reading, Badminton, Tennis, Soccer, Homelabbing, Programming

PROJECTS

Fault Tolerance and External Storage for Dynamization Systems

January 2026 – April 2026

- This project is an extension of Dr. Douglas Rumbaugh's work, [Towards Systematic Dynamic Extension of Static Indexes in VLDB 2024](#).
- I implemented fault tolerance in the form of write-ahead-logging and checkpointing. In addition to that I implemented an external ISAM tree to fulfill the data persistence requirement needed for indexes.
- This index reduced insertion time by 99.36% compared with a B-tree, with OS caching disabled for both, but point lookups were 85% slower.
- GitHub: <https://github.com/XoutDragon/dynamic-extension>

Campus Connect (Stellic Pathfinder Challenge 2026)

July 2026 – August 2026

- This project aimed to help college students and faculty connect on a familiar looking platform inspired by Discord. This platform allows communication between students and faculty, a central platform for course/club communication, marketplace, and real time updating calendar for events.
- As this project was meant to be built with AI, me and my team of three utilized Claude Code and debugged its outputs to fit the needs of our platform.
- The stack used: React, Typescript, Vite, Tailwind, and Convex

Dynamic Extension in PostgreSQL

September 2026 – Present

- This project extends Fault Tolerance and External Storage for Dynamization Systems in which I am in the process of developing a PostgreSQL extension utilizing the above framework.
- I will benchmark PostgreSQL's B-Tree with our framework using TCP-H and TCP-C and compare results.